

JVM Lab: JVM G1GC Allocation Tuning

Duration : 30 Min.

Scenario

A transaction service experiences periodic high latencies due to Stop-The-World (STW) GC pauses. You are required to optimize the Garbage Collector settings for low latency.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Tune JVM options by enabling G1GC, setting max pause target (`-XX:MaxGCPauseMillis=200`), and configuring initiating heap occupancy percent to optimize GC performance.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. G1GC policy enabling (`-XX:+UseG1GC`) and pause target configurations: 5 marks
2. Heap occupancy percentages tuning and performance logs check: 5 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.